

Strict-past joint score and heat-current reduction

TECT verification-first repository

claim A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION

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Purpose and scope

This note continues the strict-past resolvent reduction for the finite ordered Gaussian sharp-cube filtration. It keeps the positive completed square that the preceding sufficient signed-charge formulation discarded, differentiates the production coefficient increment at the conditional origin, and computes the exact conditional heat-current identity. All identities are finite-dimensional and cutoff-wise. No cutoff-uniform form bound, Nelson estimate, or measure construction is asserted.

Write $H_j = T_j + q^{-1}I$, $R_j = (I + qT_j)^{-1}$, so $H_j^{-1} = qR_j$, and let

$$C_j(u) = \frac{1}{2} \int [B(x + G_j u) - B(x)] : ((D(x + G_j u))^{\otimes 2} - \Gamma_j).$$

The strict-past coefficient score and exact nonlinear remainder are

$$\langle m_j, v \rangle = DC_j(0)[v] = \frac{1}{2} \int DB(x)[G_j v] : (Dx^{\otimes 2} - \Gamma_j), \quad N_j(u) = C_j(u) - \langle m_j, u \rangle.$$

Thus $N_j(0) = DN_j(0) = 0$.

Joint-score metric completion

For every strict-past realization and shell displacement h_j , the retained affine block plus coefficient increment satisfies the exact identity

$$\mathbb{E}_j(I_j + C_j)(\xi_j + h_j) = -\epsilon_v |h_j|^2 - \frac{q}{2} \langle \ell_j + m_j, R_j(\ell_j + m_j) \rangle + \mathbb{E}_j N_j(\xi_j + h_j) + \frac{1}{2} |H_j^{1/2}(h_j + qR_j(\ell_j + m_j))|^2.$$

Indeed, substitute $C_j(u) = \langle m_j, u \rangle + N_j(u)$ before completing the H_j -square. This is algebraic and loses no sign. On the fast relative-phase principal symbol, where $m_j = \ell_j$, the negative source rebate is four times its coefficient-one value. Hence the earlier factor-four obstruction is recovered automatically rather than inserted as a separate convention.

The preceding charge involving only ℓ_j remains a sufficient reduction, but it discards a potentially useful positive square too early. The sharper remaining object must retain the joint score, nonlinear remainder, and square together.

Conditional heat-current identity

Let $g = G_j \xi_j$, condition on the strict past, and use independence of the same-shell Gaussian value and derivative variables under the pinned entrywise-even covariance. With value covariance Σ_j and past derivative covariance Γ_{j-1} ,

$$\mathbb{E}_j C_j(\xi_j + h_j) = \frac{1}{2} \int [P_{\Sigma_j} B(x + G_j h_j) - B(x)] : (D(x + G_j h_j)^{\otimes 2} - \Gamma_{j-1}).$$

The proof expands the derivative square. The centered same-shell derivative covariance cancels $\Gamma_j - \Gamma_{j-1}$, while all value-derivative cross terms vanish. The remaining value average is precisely the Gaussian heat semigroup P_{Σ_j} .

Moreover,

$$P_{\Sigma}B(u) - B(u) = \frac{1}{2} \int_0^1 P_{t\Sigma}(\Sigma : D^2B)(u) dt.$$

This separates a translated current/Cartan contraction from Gaussian heat curvature without assigning either term an unsupported sign.

Conditional positivity of C_j is false. At $x = 0$, $Dx = 0$, and $h_j = 0$, production $B(0) = 0$ and $P_{\Sigma_j}B(0) \succeq 0$, whence

$$\mathbb{E}_j C_j = -\frac{1}{2} \text{Tr}(P_{\Sigma_j}B(0)\Gamma_{j-1}) < 0$$

for a nondegenerate past covariance. Therefore the proof may not discard the current term by a shellwise positivity claim.

Consequence and remaining gate

The next honest target is

A13-CLASSII-STRICT-PAST-JOINT-REMAINDER-CARTAN-FORM-BOUND

prove a cutoff-uniform global estimate for the sum of

$$\frac{q}{2} \langle \ell_j + m_j, R_j(\ell_j + m_j) \rangle - N_j$$

while keeping the centered Q , translated current, frame/Cartan heat curvature, and the retained positive square coupled. Termwise Young absorption or repeated payment of past energy is not admissible. Closure of this gate would feed the umbrella controlled-shell one-use estimate; it is not itself established here.

Devil's-advocate

1. **DISMISSED:** the coefficient score may be added after completing the old square. Differentiating first changes the center from ℓ_j to $\ell_j + m_j$; the primary and independent randomized identities verify the exact equality.
2. **VALID WITH MITIGATION:** same-point value and derivative noises are always independent. This holds only for the pinned entrywise-real-even covariance. A correlated synthetic control fails the heat-current identity and is included in the independent audit.
3. **UPHELD AGAINST A POSITIVITY READING:** the conditional coefficient increment is nonnegative. The origin calculation gives a strict negative production example, so no such sign is used.
4. **VALID WITH MITIGATION:** the heat-semigroup formula already bounds the global sum. It only identifies the curvature exactly; its cutoff-uniform Cartan/form estimate is the named next gate.
5. **UPHELD AGAINST PROMOTION:** this reduction proves the controlled-shell one-use or Nelson theorem. It proves neither and leaves A13 at T4.

Result footer

Result ID: A13-CLASSII-STRICT-PAST-JOINT-SCORE-HEAT-CURRENT-REDUCTION

Claim: A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION

Evidence: primary 9/9 and non-importing independent 9/9.

Verified: exact joint-score recentering, factor-four recovery, conditional heat-current identity, and two adversarial negative controls.

Tier before / after: T4 / T4 (no tier change).

No-overclaim: no global joint-remainder bound, controlled-shell one-use, Nelson theorem, interacting measure, regulator removal, or tier promotion.

Next action: prove A13-CLASSII-STRICT-PAST-JOINT-REMAINDER-CARTAN-FORM-BOUND with the positive square retained.